

Design and Simulation of a $97\ \Omega$ 16 dB Edge-Coupled Line Coupler at 64 GHz

Using ADS LineCalc and Momentum EM Simulation for RF Coupler Design and Performance Evaluation

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I. PART 1: TECHNOLOGY SELECTION AND IMPEDANCE TRANSFORMATION

A. Technology and Substrate Motivation

For the $97\ \Omega$, 16 dB edge-coupled line coupler at 64 GHz, the **Rogers 3200** substrate was chosen for its low dielectric constant ($\epsilon_r = 3.02$), moderate loss tangent ($\tan \delta = 0.002$) and copper conductivity of $5.8 \times 10^7\ \text{S/m}$, ensuring low dispersion and losses at millimeter-wave frequencies. The thickness of the board $H = 0.254\ \text{mm}$ and the thickness of the metal $T = 35\ \mu\text{m}$ were selected for the manufacturability and compactness. Using ADS LineCalc, the synthesized dimensions were as follows:

$$W = 146.1\ \mu\text{m}, \quad S = 389.8\ \mu\text{m}, \quad L = 0.783\ \text{mm},$$

yielding $Z_e = 113.8\ \Omega$ and $Z_o = 82.7\ \Omega$ for 16 dB coupling and 90° phase shift.

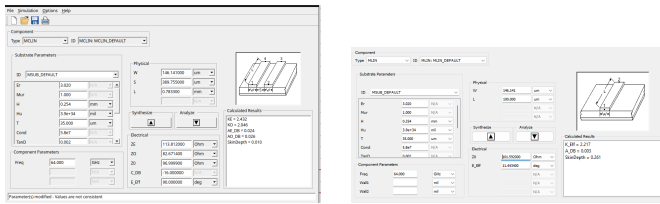


Fig. 1. Left: LineCalc synthesis window. Right: Impedance of Micro-strip line.

B. Ideal Coupled-Line Design

The ADS schematic in Fig. 1 was simulated using the defined substrate. The resulting S-parameters that confirm $S_{13} = -16.08\ \text{dB}$ and excellent matching ($S_{11} = -39\ \text{dB}$) and isolation ($S_{44} = -39\ \text{dB}$) at 64 GHz.

Overall, the Rogers 3200 implementation with integrated L-match networks achieves low loss, good isolation, and proper $50\ \Omega$ matching at 64 GHz within a compact footprint.

II. PART 2: CIRCUIT IMPLEMENTATION AND EM SIMULATION

At a frequency of 64 GHz, transmission line matching is used in the final design because it provides a lower Q value and a cleaner layout.

A. Impedance Transformation using Tlines

Comparison between lumped element and transmission line matching:

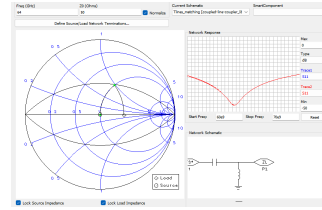


Fig. 2. Smith Chart Matching using LC

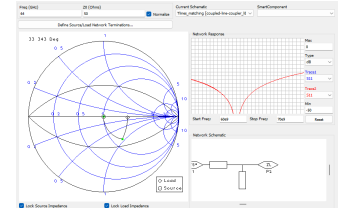


Fig. 3. Smith Chart Matching using Tlines

B. Fine Tuning and Optimization for Matching

Notice that there is a difference between the schematic result and EM simulation. The ADS optimizer was used to adjust the length of the matching lines.

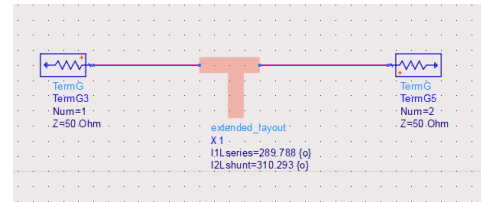


Fig. 4. Tuning Process

C. Adaptation to Realistic Layout

A realistic schematic and a realistic layout are shown below.

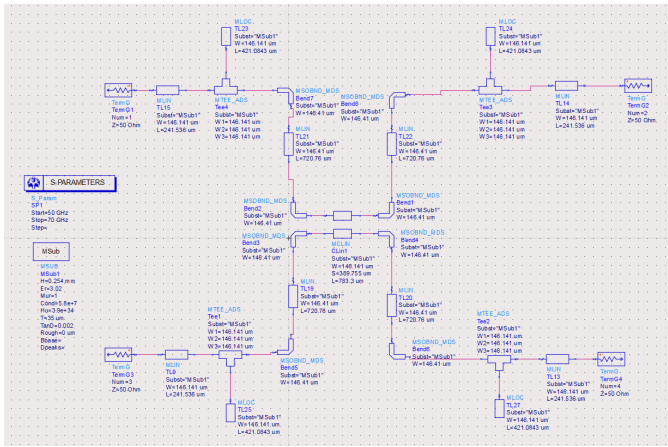


Fig. 5. Schematic of the realistic circuit

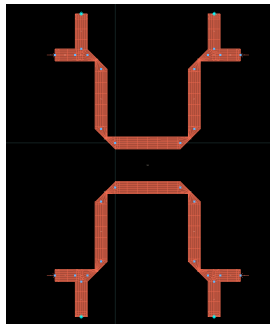


Fig. 6. Layout of the realistic circuit

III. PART 3: RESULTS

A. Matching Network Performance

The ideal schematic and the EM simulation never perform the same. For the ideal schematic shown in Fig. 7, S_{11} has a magnitude of 0.003 in the schematic and 0.107 in the EM simulated result. For the ideal layout shown in Fig. 8, S_{11} has a magnitude of 0.156 in the schematic and 0.024 in the EM simulated result.

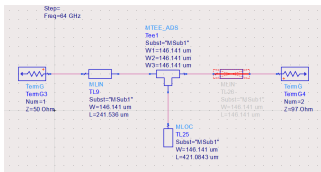


Fig. 7. Case for Ideal Schematic Matching

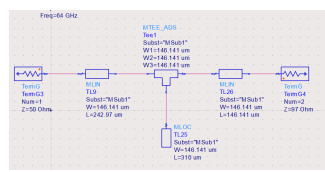


Fig. 8. Case for Ideal Layout Matching

B. Results from Ideal Schematic and its Layout

- 1) Impedance Matching: See Fig.9 and Fig.10
- 2) Insertion Loss: See Fig.11 and Fig.12

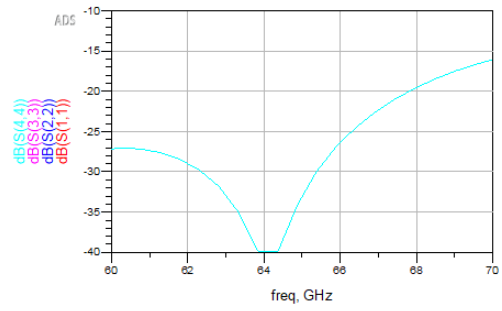


Fig. 9. Impedance Matching of the Ideal Schematic

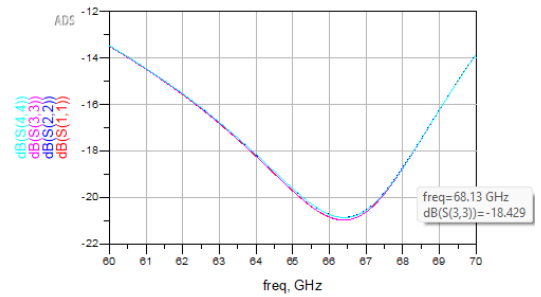


Fig. 10. Impedance Matching of the Layout

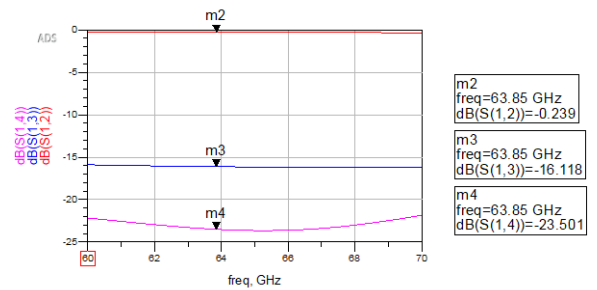


Fig. 11. Insertion Loss of the Ideal Schematic

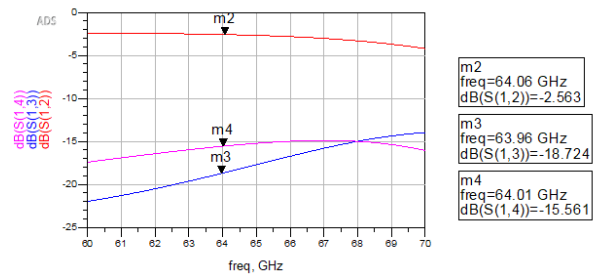


Fig. 12. Insertion Loss of the Layout